



Reviewing the SmartPLS 4 software: the latest features and enhancements

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Abstract

Partial least squares structural equation modeling (PLS-SEM) is a highly popular multivariate data analysis method. The SmartPLS 3 software program helped many marketing researchers analyze the complex relationships between latent variables (i.e., mediation, moderation, etc.), which they measured by means of sets of observed variables. This program's intuitive graphical user interface and various features, such as new metrics (e.g., HTMT, model fit indexes), advanced techniques (multigroup analysis, PLS_{predict}), and complementary techniques (e.g., confirmatory tetrad analysis, importance-performance map analysis), which impacted many business disciplines. SmartPLS 4 represents a significant leap forward in development with its completely revamped graphical user interface, faster processing speed for data estimation, and new model assessment features (i.e., cross-validated predictive ability test, endogeneity assessment, and a necessary condition analysis). This paper reviews SmartPLS 4 and discusses its various features, thereby providing researchers with concrete guidance that fits their analytical research goals.

Keywords Partial least squares structural equation modeling · Partial least squares path modeling · PLS-SEM · Composite modeling · SmartPLS · Multivariate data analysis

Introduction

Partial least squares structural equation modeling (PLS-SEM; Wold 1982; Chin 1998) also known as PLS path modeling (Tenenhaus et al. 2005; Rigdon 2012), is a highly popular multivariate data analysis method used in business and economics, social sciences, and in other fields such as agriculture, ecology, environmental sciences, engineering, geography, medicine, and psychology (Cheah et al. 2023; Guenther et al. 2023; Sarstedt et al. 2022; Magno et al. 2022;

Wang et al. 2023). Using the keywords “PLS-SEM” and “PLS path modeling” (setting “articles” as the document type, “English” as the language, and “2015¹–2022” as the publication window) to search the Web of Science database, showed a remarkable upward trend in the use of PLS-SEM (see Fig. 1). Using the keywords “SmartPLS” and “Smart PLS”² with the same settings, revealed a similar upward trend, suggesting that PLS-SEM's growing popularity and, specifically, that of the SmartPLS software program (Ringle et al. 2005, 2015, 2022) are closely related. It is clear that the availability of a program with a user-friendly graphical user interface (GUI) played an important role in the dissemination of PLS-SEM (Sarstedt and Cheah 2019).

Sarstedt and Cheah (2019) outlined that PLS-SEM's growing popularity began with Chin (1998), Tenenhaus et al. (2005), and Hair et al.'s (2011) seminal articles. SmartPLS 2's release (Ringle et al. 2005), which changed the PLS-SEM software landscape from pure syntax to a proprietary software program with a GUI, certainly facilitated

¹ SmartPLS 3 was released in 2015, contributing to the wide-scale adoption of PLS-SEM (Sarstedt and Cheah 2019).

² Even though the software's correct name is SmartPLS, we also included the keyword “Smart PLS,” as this label is used in a remarkable number of published articles.

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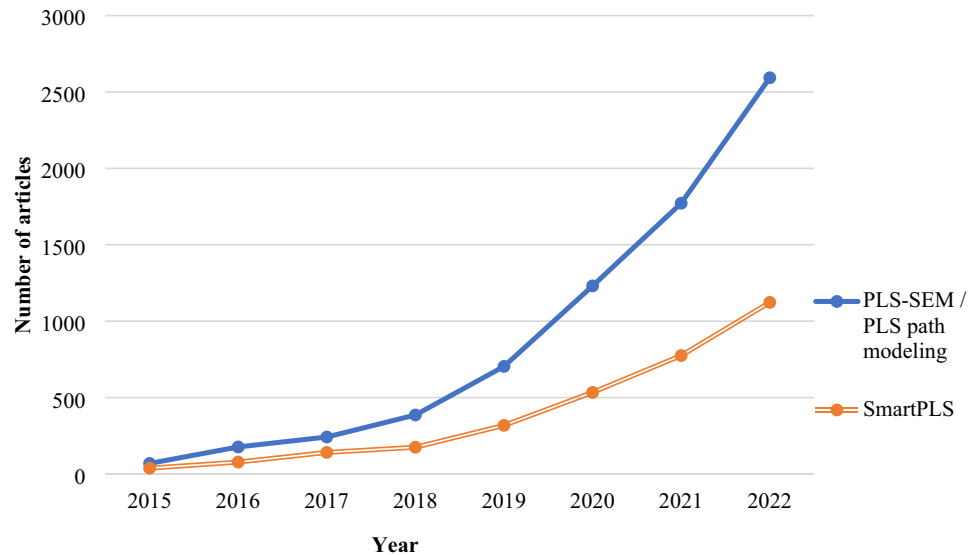
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Fig. 1 Number of articles using the PLS-SEM (or PLS path modeling) technique and the SmartPLS software



its popularity (Sarstedt and Cheah 2019). This program—characterized by distinctive usability features for undertaking standard PLS-SEM analyses—was initially available for free. In 2015, Ringle et al. (2015) introduced SmartPLS 3, a commercial package providing a far broader range of modeling and algorithmic options, improved usability, and professional support. Furthermore, it was compatible with both the Microsoft and Mac operating systems.

Seven years later, Ringle et al. (2022) upgraded the entire program and released version 4, which comes with various improvements and features, such as a redesigned GUI, an improved performance and data import, new algorithms, and additional statistical routines beyond PLS-SEM.

This article aims to help PLS-SEM users discover the above benefits to allow them to fully utilize SmartPLS 4's potential. Since Sarstedt and Cheah (2019) already provided a review of established SmartPLS features, this article focuses on and reviews its new features to ensure that researchers are fully acquainted with this software's potential. This review's aim is to help researchers make informed decisions regarding their tool selection, data analysis, result reporting, and the interpretation of their findings when using SmartPLS 4's new features.

The SmartPLS 4 software

The new core features

One of the most important of SmartPLS 4's new core features is its improved data import. SmartPLS 4 no longer only relies on comma-separated value (CSV) files to import data into the program; currently, it reads data files directly from IBM® SPSS® and Excel® (Sarstedt and Mooi 2019)

without requiring prior conversion to a CSV file, which provides users with more data entry options. Furthermore, existing SmartPLS 3 projects can be imported into and updated in SmartPLS 4. After importing a data file, users can define each observed variable's scale type as metric, ordinal, categorical, and binary,³ as well as change the indicator names using the new setup option in the data view. Many new PLS-SEM sample projects have also been added, including employee retention (Hair et al. 2018), the technology acceptance model (TAM; Anderson et al. 2011), and the unified theory of the acceptance and use of technology (UTAUT; Al-Gahtani et al. 2007), all of which complement the previously available models and data files on corporate reputation (Schwaiger 2004; Sarstedt et al. 2023a, b) as well as the European Customer Satisfaction Index (ECSI; Eklöf and Westlund 2002). All of the new sample projects serve as showcase models that allow users to employ the corresponding algorithms and various PLS-SEM techniques proficiently to reproduce published literature results. Importantly, users can also save their results reports within SmartPLS 4 (or export them to Excel®) for better transparency and reproducibility.

SmartPLS 4 has a new GUI with an intuitive look, which allows users to specify and estimate their path models easily.

³ A binary variable is a categorical variable with only two possible values (e.g., yes/no), which a dummy-coded (0/1) variable commonly represents (Sarstedt and Mooi 2019). As Becker et al. (2023) outline, the results that SmartPLS 4 provides in respect of such moderator effects can subsequently be interpreted to obtain meaningful insights. Specifically, SmartPLS 4 allows binary variables to not be standardized; consequently, researchers are not required to make manual corrections to interpret the interaction effects involving binary variables (Becker et al. 2023).



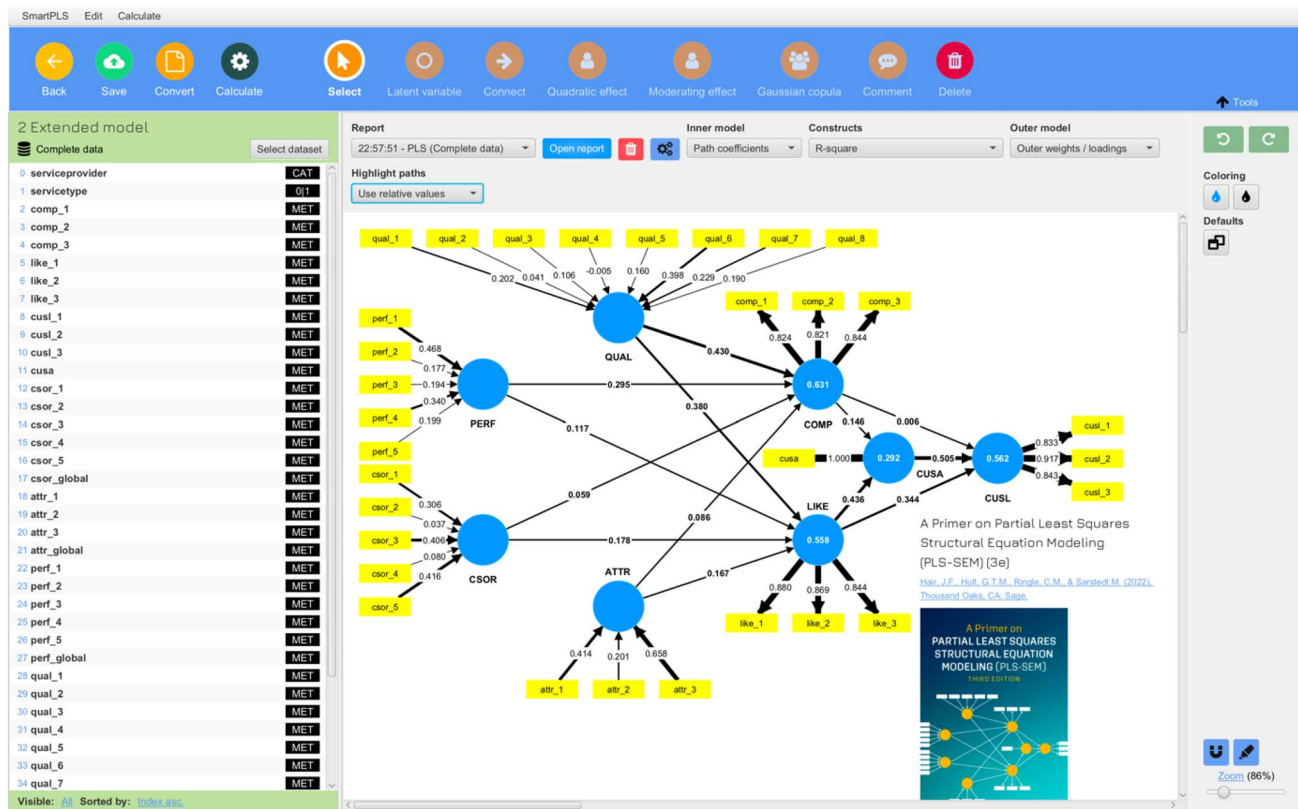


Fig. 2 The SmartPLS 4 GUI with model estimates

For example, users can create new constructs by dragging and dropping selected items from the indicator menu to the modeling window. Thereafter, the constructs can be shaped into circles, rectangles, hexagons, or octagons, while the weighting modes can be specified as Automatic, Mode A, Mode B, Equal Weights, or Pre-Defined. Furthermore, the constructs can be linked by drawing single-headed arrows between them, or using dotted, dashed, solid straight or solid curved lines. The new version also incorporates extra languages, including Croatian, Norwegian, Swedish, Thai, and Vietnamese.

When estimating path models, as shown in Fig. 2, the new version allows users to select the “recalculate automatically” option, which automatically re-runs the estimations after items, constructs or path relationships have been modified and/or deleted, and after changes have been made to the data file. This helpful function allows users to quickly estimate and assess the measurement and structural models, thereby aligning with Wold’s (1985, p. 590) notion that PLS-SEM facilitates “a dialogue between the investigator and the computer.”

Furthermore, users can now compare results when estimating different models or when relying on different algorithms (i.e., the standard PLS-SEM algorithm or the consistent PLS-SEM algorithm), weighting schemes (i.e., factor,

path, or principal component analysis), types of results (i.e., whether unstandardized, standardized, or mean-centered), or types of bootstrapping confidence intervals [i.e., percentile, studentized, or bias-corrected and accelerated (BCa)]. Another useful new feature is that users do not need to save the latent variable scores from the PLS-SEM algorithm estimation to a separate data file before proceeding to the second stage of estimating higher-order constructs (i.e., executing the embedded or disjoint two-stage approaches; Becker et al. 2023; Sarstedt et al. 2019).

The new GUI allows PLS-SEM users to display moderating and quadratic effects more intuitively (Becker et al. 2023; Hair et al. 2024), eliminating the need to generate the interaction terms as in SmartPLS 3. In other words, SmartPLS 4 represents the conceptual model by drawing single-headed arrows pointing from the moderator to the direct effect between the constructs, whereas SmartPLS 3 showcased the statistical model, therefore showing the interaction term. However, in the moderator analysis’s results report, which SmartPLS 4 provides, users find the interaction term and the relationship between the moderator and moderated relationship’s dependent variable, which the software adds automatically.

In SmartPLS 3, the interaction term was generated using either the product indicator approach, the orthogonalizing



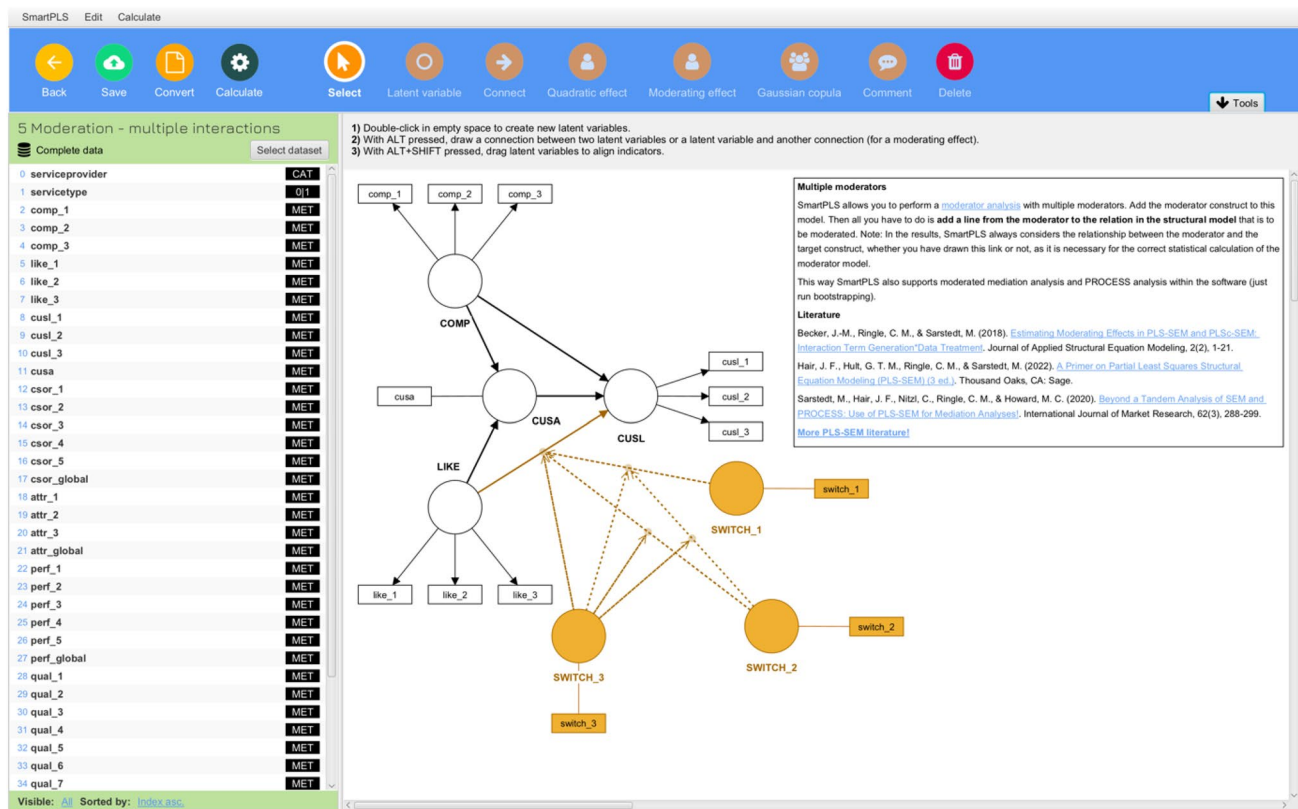


Fig. 3 The SmartPLS 4 GUI with three-way interactions

approach, or the two-stage approach, whereas SmartPLS 4 always uses the two-stage approach. This update is rooted in well-documented PLS-SEM literature indicating that the latter approach simplifies the modeling process, increases the statistical power, improves the interpretability, and facilitates the model assessment (Becker et al. 2018, 2023). In addition, SmartPLS 4 enables users to implement three-way (or higher) interactions effortlessly. Figure 3 is a snapshot of the GUI that includes a moderator analysis with multiple moderators from the corporate reputation model.

SmartPLS 4 enhances the out-of-sample predictive accuracy assessment by incorporating the cross-validated predictive ability test (CVPAT; Sharma et al. 2023a; Liengaard et al. 2021). This test is an alternative to $PLS_{predict}$ (Shmueli et al. 2016, 2019) for the prediction-oriented assessment of PLS-SEM results and can be executed by clicking on “PLSpredict/CVPAT” under the “calculate” menu. The test helps researchers compare a model statistically with a naïve indicator-averages prediction benchmark as well as with a more demanding linear model prediction benchmark in terms of its out-of-sample predictive accuracy, which is based on an inferential test. In order to achieve this assessment, PLS-SEM’s average loss value is compared to that of a prediction using indicator averages (IA) as a naïve benchmark as well as a linear model (LM) estimation as

a more conservative benchmark. Strong predictive validity is revealed if PLS-SEM’s average loss is lower than that of the benchmarks, which a negative difference in the average loss values conveys (Sharma et al. 2023a). The CVPAT tests whether PLS-SEM’s average loss is significantly lower than that of the benchmarks; consequently, the difference between the average loss values should be significantly below zero (p value < 0.05) to substantiate the model’s better predictive capabilities when compared to those of the prediction benchmarks (see Fig. 4). Moreover, this test could be used to evaluate either a single key target endogenous construct or, simultaneously, multiple relevant target constructs. As pointed out by Chin et al. (2020), the software therefore fulfills PLS-SEM’s original purpose of enabling users to consider their modeling’s causal-predictive nature.

Finally, SmartPLS 4 incorporates the Gaussian copula approach (Park and Gupta 2012), one of the new features designed to detect and address endogeneity issues in the PLS-SEM (Hult et al. 2018). This approach is known as the instrumental variable-free approach, which allows researchers to control for endogeneity by directly modeling the correlation between the endogenous variable and the error term by means of a copula (Hult et al. 2018). The Gaussian copula control function might also apply to the numerous endogenous regressors, which necessitate the simultaneous



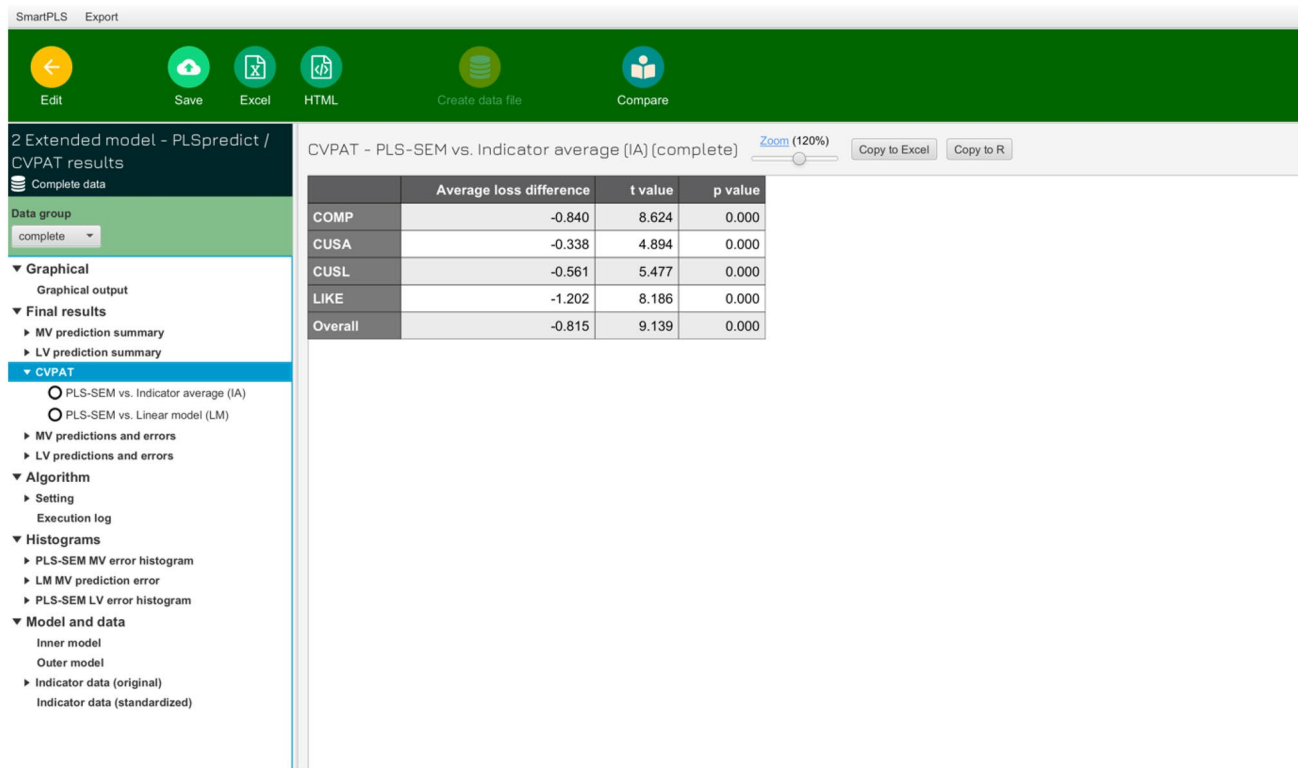


Fig. 4 A CVPAT result

inclusion of several copula terms, namely, one for each regressor (see Fig. 5). The parameter estimate for this copula term is the estimated correlation between the regressor and the error term scaled by the error's variance. A statistical test of this parameter estimate, using bootstrapped standard errors, should be able to determine whether this association is statistically significant and to detect an existing endogeneity issue (Hult et al. 2018; Sarstedt et al. 2020b). However, researchers need to consider the Gaussian copula approach's requirements and limitations (Park and Gupta 2012; Becker et al. 2022). The nonnormality of the endogenous variable(s) is a critical condition for the Gaussian copula technique to be implemented (Park and Gupta 2012). Researchers may use the Cramer-von-Mises nonnormality test, which is available in SmartPLS 4, to determine the nonnormality of the endogenous variable(s).

Additional statistical routines beyond PLS-SEM

SmartPLS 4 has introduced several new statistical routines aimed at offering users an opportunity to create and estimate a model beyond the PLS-SEM method. These include a regression model (multiple and logistic), path analysis (i.e., regression analysis with multiple dependent indicators), PROCESS (Hayes 2022; Sarstedt et al. 2020a) to handle conditional mediation analysis (also known as moderated

mediation; see Cheah et al. 2021), and a necessary condition analysis (NCA; Dul 2016; Richter et al. 2020), which, based on a necessity logic, allows users to determine the factors that need to be in place for a desired outcome to occur (Richter et al. 2020). The reason for including these features is to provide researchers with a platform for estimating their research in a one-stop approach, such as by conducting a multi-method analysis or by comparing the results across analysis approaches. With this in mind, we anticipate that the upcoming versions of SmartPLS 4 will include further interesting features, such as generalized structured component analysis (GSCA; Hwang and Takane 2004; Hwang et al. 2023) as one of the composite-based structural equation modeling techniques.

Covariance-based structural equation modeling (CB-SEM; Jöreskog 1978; Rigdon 1998) is a particularly promising new SmartPLS 4 feature, which allows researchers to create, estimate, and evaluate models using CB-SEM with a GUI that is far easier to navigate than IBM® SPSS® Amos when, for example, running and modifying a confirmatory factor analysis model. This feature allows researchers to conduct a comparison between results estimated by PLS-SEM and CB-SEM algorithms (see Fig. 6), and to convert an existing model type into a new model type by, for example, converting an existing PLS-SEM model into a CB-SEM model by simply clicking the “Convert” button on the main



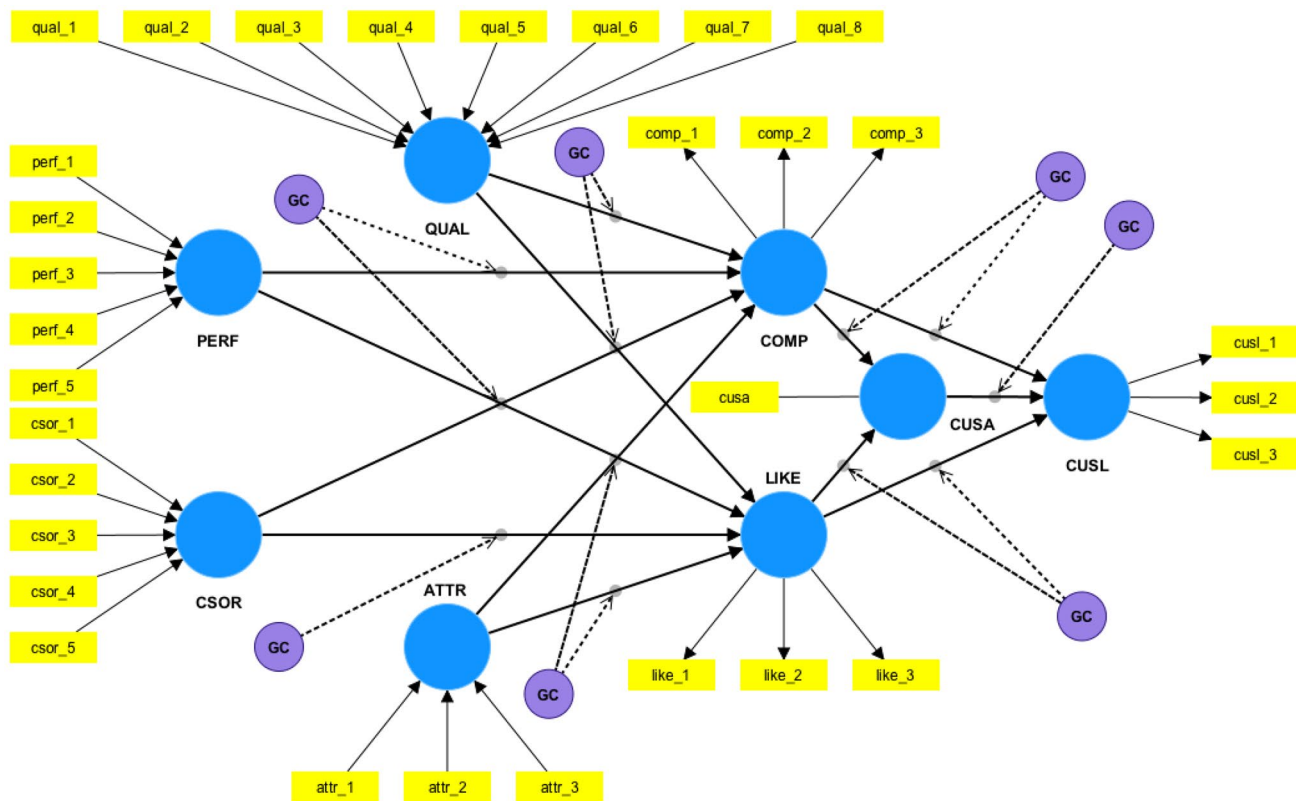


Fig. 5 SmartPLS 4 GUI with Gaussian copulas to address endogeneity issues

toolbar. Furthermore, SmartPLS 4 enables increasingly sophisticated CB-SEM analyses, including latent growth model estimates for longitudinal studies (see Fig. 7), Mardia's index computation for multivariate normality, and the root mean square error of approximation (RMSEA) for confidence interval estimation. Moreover, we anticipate further CB-SEM function advances in SmartPLS 4, which should include multi-group analysis and similar extensions that users often apply. Consequently, SmartPLS' CB-SEM capabilities should quickly equal the rich methods portfolio that software applications, such as Mplus or the lavaan package of the R statistical programming language, offer.

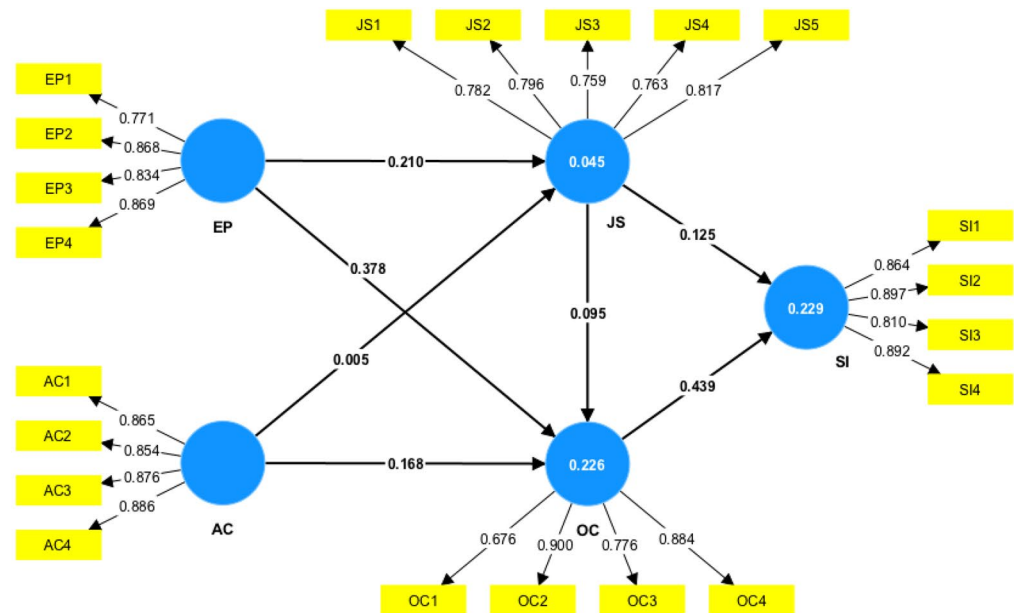
Given the recent methodological additions to the SmartPLS software, including CB-SEM, there may be a concern about potential brand dilution or confusion. It should be noted that the name "SmartPLS" has traditionally been associated with PLS-SEM, and the inclusion of CB-SEM functionality may come as a surprise to some users. Researchers and practitioners who rely on SmartPLS for PLS-SEM analysis should be aware of these expanded capabilities, which may necessitate some adaptation and careful consideration of the specific methodology they intend to use. While SmartPLS remains an important tool for PLS-SEM, users should consider these recent developments and determine whether they meet their analytical requirements.

With great power comes great responsibility

Compared to certain covariance-based SEM tools, such as LISREL and EQS, which rely on linear algebra or formulas to specify measurement and structural models, a GUI-based program like SmartPLS 4 facilitates PLS-SEM users' learning process when analyzing their research models. However, there is a concern that the PLS-SEM reporting process might be misused as a mechanistic, opaque, and manipulative approach by those wanting to achieve specific results. As indicated by Petter and Hadavi (2021), users should take care to interpret PLS-SEM data accurately and to explain the findings to their audiences. Indeed, Petter (2018) makes the point that controversies will continue if PLS-SEM's power does not lead to appropriate use responsibility (i.e., adhering to PLS-SEM best practices) and if the chosen approach is not clearly justified (i.e., dealing with formatively measured constructs, assessing a model from a prediction perspective, etc.). For example, certain SEM scholars have advocated abandoning the PLS-SEM method in research publications, claiming that it produces biased and inconsistent parameter estimates (e.g., Rönkkö et al. 2016), even though others have extensively countered this criticism (Cook and Forzani 2023; Henseler et al. 2014; Russon and Stol 2023; Sharma



PLS-SEM



CB-SEM

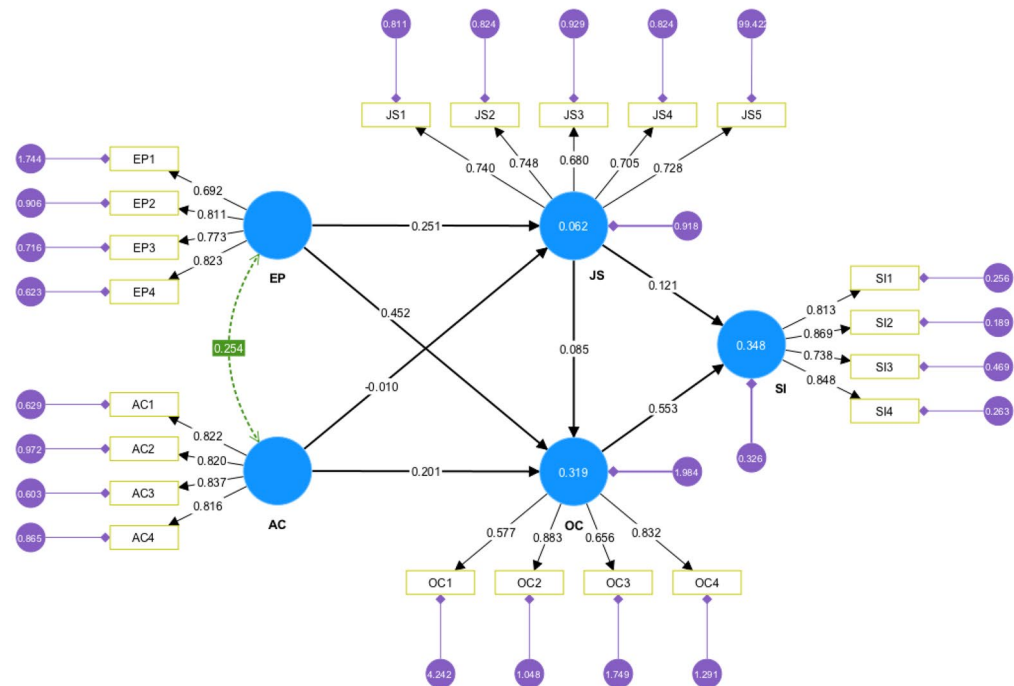


Fig. 6 Comparison between PLS-SEM and CB-SEM results generated by SmartPLS 4

et al. 2023b; Sarstedt et al. 2016; Hair et al. 2019)—see also Petter (2018) and (Petter and Hadavi 2021).

These debates on the PLS-SEM method might frustrate and discourage some users, but they help us to reconsider and rethink how to use SmartPLS 4 wisely, particularly when engaging with the reporting procedure. For example, Hair et al. (2012) and Sarstedt et al. (2022) emphasize the need to be transparent when reporting PLS-SEM analyses' results to facilitate their replicability, starting with reporting

the PLS algorithm settings. Such transparency will help researchers, reviewers, and editors assess PLS-SEM's suitability for a specific purpose and when evaluating its results.

We next present guidelines for making informed decisions when selecting from SmartPLS 4's various settings, and when reporting them transparently.

First, regarding PLS algorithm settings, there are three common types of structural model weighting schemes: the factor weighting scheme, path weighting scheme, and



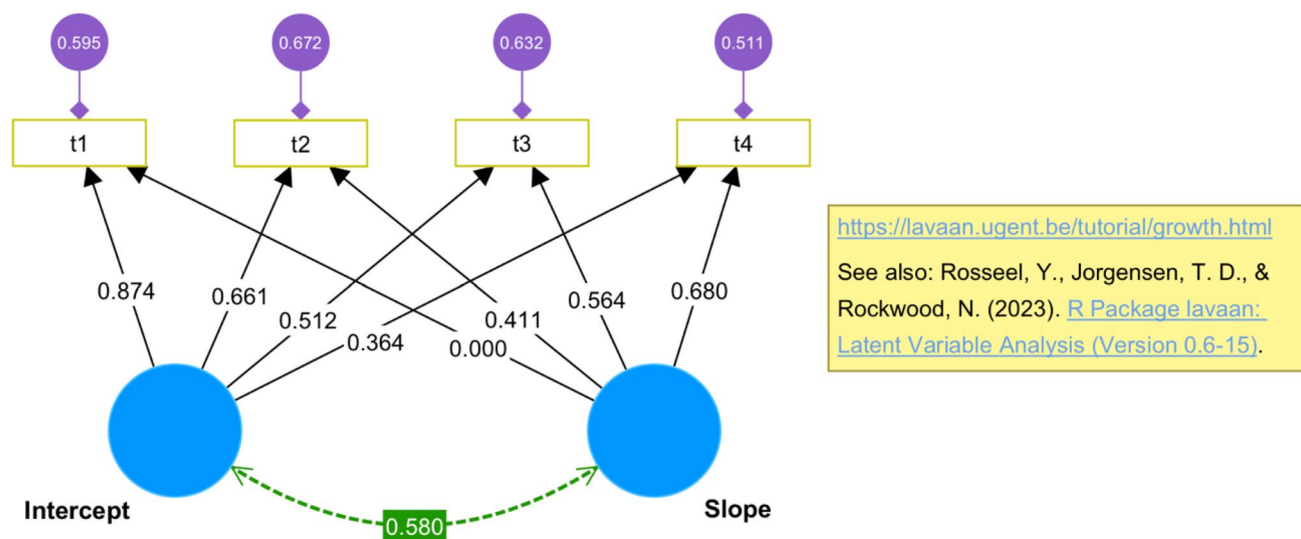


Fig. 7 Latent growth model (LGM) examples in SmartPLS 4

centroid weighting (Wold 1966; Chin 1995). The centroid method is an older approach suggested by Wold (1982), which, however, produces unstable results of constructs correlations that are close to zero (see Henseler et al. 2009). Because of this limitation, the centroid weighting scheme has been abandoned in SmartPLS 4. While the results for the factor and path weighting schemes differ little, path weighting is generally recommended as it produces the highest R^2 value and is applicable to all kinds of PLS path model specifications and estimations (Hair et al. 2022). In addition, researchers can use the principal component analysis (PCA) option to estimate the model, which has been newly included in SmartPLS. This weighting scheme applies a PCA per construct (Sarstedt and Mooi 2019) and uses the resulting score vectors as input for structural model regressions. This weighting scheme thereby helps to mitigate multicollinearity by creating orthogonal (uncorrelated) components, making it easier to estimate model parameters and reducing issues related to multicollinearity. We recommend reporting the selected weighting scheme, together with the maximum number of iterations (which, in SmartPLS 4, is a default of 3000) and the stop criterion (which is a default of 10^{-7}) when running the PLS-SEM algorithm (see Hair et al. 2022).

Second, all studies should clearly report the mode used to compute the composites, specifically the choice between Mode A and Mode B. While this point has been well documented in the PLS-SEM literature, most users tend to overlook the reporting of the estimation mode, resulting in an ongoing heated debate on the PLS-SEM technique's potential bias. By clicking on a latent variable, SmartPLS 4 users can specify either Mode A or Mode B (or, alternatively, equal weights or pre-defined weights). In the case of a reflective measurement model, SmartPLS 4 by default computes the

composites using Mode A (i.e., correlation weights), while, in the case of a formative measurement model, it uses Mode B (i.e., regression weights). However, SmartPLS 4 users should be aware that the software always computes both the loadings and the regression weights, regardless of the mode choice, while the loadings are used in Mode A, but not in Mode B (and vice versa for regression weights). Alternatively, researchers interested in estimating a common factor model's results, should realize that SmartPLS 4 provides an option to run the PLS algorithm to mimic common factor model results (Dijkstra and Henseler 2015).

Third, users should clearly report the bootstrapping setting when evaluating a structural model, including the number of bootstrap subsamples, the test type (one-tailed vs. two-tailed), the significance level, the random number generator (random seed vs. fixed seed), and the method used to calculate the confidence interval. Regarding the number of bootstrap subsamples, Becker et al. (2023) propose that, while the initial model estimations could be based on a small number of subsamples to save time (e.g., 1000), the final analysis should use at least 10,000 subsamples to ensure the results' stability (see also Streukens and Leroi-Werelds 2016). In terms of the test type and the significance level, it is essential for a one-tailed or two-tailed test's results to be reported and based on the way researchers posit their hypotheses, as well as at a significance level of 10% ($\alpha = 0.10$), 5% ($\alpha = 0.05$), or 1% ($\alpha = 0.01$). Rather than relying on the random seed option, the fixed seed option could be used in the analysis to avoid concerns about the results' replicability, which might arise from p-hacking or p-harking (Sarstedt and Adler 2023). This is a new option available in SmartPLS 4, which produces the same results if the same number of subsamples is drawn. Three bootstrapping methods are available



to estimate nonparametric confidence intervals: percentile bootstrap, studentized bootstrap, and BCa bootstrap. These methods differ in how they handle the statistic of interest's sampling distribution, as well as in their ability to correct bias and account for distributional characteristics. A suitable bootstrapping method should be selected and the results reported accordingly, given that each of these approaches has its own properties (Aguirre-Urreta and Rönkkö, 2018; Hayes and Scharkow 2013). Following Becker et al. (2023), we recommend using percentile bootstrapping, which is SmartPLS 4's default setting, although researchers should use the BCa approach when there is a highly asymmetric parameter distribution (e.g., detected by inspecting the histogram of a parameter's bootstrap distribution).

Finally, given the growing concerns about research results' replicability, the reporting of descriptive statistics (i.e., the mean and the standard deviations) and of indicators' matrixes and construct-level correlations should be mandatory. It is difficult for researchers to replicate findings and to determine whether model assessment measures are valid in the absence of such results (Adler et al. 2023a). It could be argued that including such information within a study's main text means that researchers risk exceeding the maximum number of allowed words, which is why we encourage authors to include these additional findings in the appendix as supplementary materials, or to include a footnote indicating a specific site where more information can be found. In fact, Adler et al. (2023b) have recently proposed a preregistration template specifically for PLS-SEM-based studies to engage in open science practices. The purpose of adhering to such reporting procedures and open science practices is to increase the confidence of both reviewers and readers in the findings of studies using the PLS-SEM technique, such as increased transparency and reproducibility.

Conclusion

SmartPLS remains the most comprehensive software for performing PLS-SEM analyses and is business field researchers' preferred choice due to its user-friendly interface, ease of use, and continuous introduction of innovative features. SmartPLS developers release regular updates that not only fix bugs, but also expand the software's features beyond those available in competing PLS-SEM packages, such as ADANCO, PLS-Graph, and WarpPLS. Although PLS-SEM users can access the free PLS-SEM modules in R or Stata (Hair et al. 2021; Monecke and Leisch 2012; Rademaker and Schuberth 2021; Venturini and Mehmetoglu 2019), these come with extensive documentation and are targeted at more experienced users with profound programming knowledge. SmartPLS 4, with all its new features, meets researchers' expectations in terms of performing advanced analyses in a

timely manner, while the new GUI not only allows them to quickly specify their PLS path models, but also empowers them to conduct more advanced and robust analyses.

Textbooks that introduce the PLS-SEM method using the SmartPLS 4 software (Hair et al. 2022; Chua 2023), some of which have been translated into multiple languages, including German, Spanish, and Vietnamese, continue to facilitate the adoption of this software and the application of advanced PLS-SEM techniques (Hair et al. 2024).

Overall, the frictionless design of SmartPLS 4 outperforms SmartPLS 3, making the implementations of PLS-SEM easier for researchers from a wide variety of disciplines. However, when employing the PLS-SEM technique and the software, researchers should exercise caution and act responsibly by following the best practices.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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